02LQD Specification and Simulation of Digital Systems

2019/06/21

1. Combinational circuit design (5 points)

An unsigned number on 4 bits $n_3n_2n_1n_0$ is the input to a prime number detector circuit. The output z is 1 if and only if the input is a prime number. Design in VHDL a combinational circuit implementing the function above. Steps: write the truth table, the Karnaugh map, minimize the 2-level Boolean expressions, describe the circuit in VHDL resorting to the 2-level structural description style.

2. FSM design (12 points)

A synchronous sequential circuit has 3 inputs *Reset, Start* and *Stop* and 1 output *Run*. The duration of *Start* and *Stop* is not known a priori, i.e. it could span several clock cycles. When *Start* becomes 1, no matter the value of *Stop*, *Run* becomes 1. *Run* remains 1 until *Stop* becomes 1, no matter the value of *Start*, then *Run* goes to 0. Changes on *Run* are synchronous with the rising edge of the clock. Reset is asynchronous.

1. Draw the state diagram of the Moore machine (**3 points**)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (**3 points**)
3. Design the FSM resorting to the VHDL behavioural description style with 1 process. The timing behavior may differ from the one of step 2 (**3 points**)
4. Design the FSM resorting to the VHDL behavioural description style with 1 process. The timing behavior must be the same of step 2. (**3 points**)

3. HLSM design (13 points)

Let X be a pure binary number on 2^n bits. Let L be its integer logarithm in base 2. Remember: L is the biggest integer such that $2^L \leq X$. Example: if $X = 312$, $L = \log_2(X) = 8$ as $2^8 \leq 312$ but $2^9 > 312$. Let *Start* be the input signal that starts computation when it is 1 and let *Done* be the output signal that tells the computation is over when it is 1. L is available at each clock tick, but it makes sense only when *Done* is 1.

Design a HLSM to compute the base-2 integer logarithm L of its input X when *Start* rises to 1 and n=4. *Done* set to 1 signals that a proper value of L is available on the output. Reset is synchronous. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior.

Exam rules:

- upload on Portale della didattica in section Elaborati:
  - running VHDL code and testbench
  - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

  no later than Monday June 24 7pm
- remember: no upload ⇒ no correction ⇒ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.